SecureLab

User Manual

RSA + AES Encrypted Hospital Lab Report Portal

Portal	Who it is for	Section
Patient	People who receive and read lab reports	2
Laboratory staff	Technicians who upload reports	3
Administrator	People who manage staff and review security logs	4

Important: This is an academic prototype. Use only synthetic (made-up) data. Never upload real patient information.

1. Getting started

SecureLab lets laboratory staff upload PDF lab reports that are **encrypted** before they are saved. Only the patient who owns a report can unlock and read it, using a private key file that stays on the patient's own computer. Even administrators and lab staff cannot read a patient's report after upload.

1.1 Starting the website (first time)

The site has two parts that must both be running. Open two terminal windows in the **SecureLab** folder.

1	**Terminal 1 (backend):** `cd backend`, then `.\.venv\Scripts\Activate.ps1`, then `python run.py`. (First time only, also run `flask --app run.py seed-demo` to create the admin and lab accounts.)
2	**Terminal 2 (frontend):** `cd frontend`, then `npm run dev`.
3	Open **http://127.0.0.1:5173** in a modern browser (Chrome, Edge or Firefox).
4	To check the backend, visit **http://127.0.0.1:5000/api/health**. It should show status "ok".

1.2 Demo accounts

Role	Email	Password
Administrator	admin@securelab.local	AdminDemo!2026
Laboratory staff	lab@securelab.local	LabDemo!2026
Patient	Create your own (see Section 2.1)	At least 10 characters

1.3 The login screen

- **Sign in** - enter email and password. If two-factor authentication is on, a second screen asks for a 6-digit code.
- **Create an account** - the link below the form. This creates a *patient* account only; staff accounts are made by an administrator.
- Every role sees a left sidebar with its menu, a **Sign out** button at the bottom of the sidebar, and a two-factor card at the top of each page.

1.4 Two-factor authentication (all roles, optional)

The card titled **Two-factor authentication** appears at the top of every page. It adds a 6-digit code from an authenticator app to your login.

1	Click **Set up authenticator**.
2	Copy the **Authenticator secret** shown, and add it to an authenticator app (Google or Microsoft Authenticator) using "enter a setup key".
3	Type the current 6-digit code and click **Confirm MFA**. The card shows **ENABLED**.
4	From now on, signing in asks for the code after your password.

Important: Keep your authenticator app. If you lose it you cannot sign in to that account in this prototype.

Academic demo - use synthetic data only

2. Patient portal

Patients register themselves, create an encryption key once, and then decrypt their reports whenever staff upload them.

2.1 Register

1. On the login screen click **Create an account**.
2. Fill in **Full name**, **Email address**, **Password** (at least 10 characters). **Patient ID** is optional; leave it blank to have one generated.
3. Click **Create patient account**. You are signed in automatically.

2.2 One-time setup: create your encryption key

Until you do this, lab staff **cannot** send you reports. The dashboard shows **Encryption key: Not set up**.

1. In the box **Generate your patient encryption key**, type a strong passphrase of **12 or more characters**.
2. Click **Generate key pair**.
3. Your browser downloads a file named like `securelab-PATIENTID-private-key.securelabkey`. The status changes to **Registered**.

> **Important:** This file and its passphrase are the **only** way to read your reports. The server has no copy. Back the file up somewhere safe and never share the passphrase. If either is lost, your reports cannot be recovered.

2.3 Unlock your key (every time you sign in)

Your unlocked key lives only in browser memory. Reloading the page or signing out locks it again.

1. In the **Patient key vault** box, click **Encrypted key file** and choose your `.securelabkey` file.
2. Enter your **Key-file passphrase**.
3. Click **Unlock private key**. The Private key card changes to **Unlocked**.

2.4 Read a report

1. Scroll to **Your laboratory reports**. Each report card shows its title, test type, date and an **ENCRYPTED** label. Click **Refresh** if a new report has not appeared.
2. Click **Decrypt & verify report** on a card (the button says "Unlock key to view" until your key is unlocked).
3. The browser decrypts the report and checks it three ways: the encryption authentication tag, the SHA-256 fingerprint, and the hospital's digital signature.

4	If all checks pass, the PDF opens in a new tab and a green message confirms it was verified. If any check fails, the PDF is **not** opened and a red error is shown.

> **Tip:** If the PDF does not appear, allow pop-ups for this site in your browser.

2.5 Dashboard cards at a glance

Card	Meaning
Available reports	How many reports have been uploaded for you
Encryption key	Registered = ready to receive reports; Not set up = do Section 2.2
Private key	Unlocked = you can open reports; Locked = do Section 2.3

Academic demo - use synthetic data only

3. Laboratory staff portal

Sign in with a laboratory account (demo: lab@securelab.local). The sidebar has two pages: **Upload report** and **Uploaded reports**.

3.1 Upload a report

Before you start, the patient must already have registered **and** created their encryption key. Otherwise they do not appear in the patient list.

1	Open **Upload report**.
2	Choose the person in **Patient account**. To find them, type a name, email or patient ID under **Search patients** and click **Search**. A green line confirms "RSA public key registered".
3	Enter a **Report title**, for example "Complete Blood Count - October 2026".
4	Pick a **Test category** (Complete Blood Count, Lipid Profile, Blood Glucose, Liver, Kidney, Thyroid, Urine Analysis or Other).
5	Click the **PDF report** box and choose a PDF file (PDF only, 12 MB maximum). A synthetic sample is provided at `docs/synthetic-demo-lab-report.pdf`.
6	Click **Encrypt & upload report**. A green message confirms the report was encrypted and stored.

What happens behind the scenes: the server encrypts the PDF with a fresh AES-256 key, locks that key with the patient's RSA public key, signs the original with the hospital key, and saves only the encrypted data.

3.2 Uploaded reports

This page lists every report your account uploaded: title, patient, test type, upload time, and an **Encrypted** status. Staff cannot view the original PDF after upload. That is by design.

3.3 Common problems

Message / symptom	What to do
Patient is missing from the list	Ask the patient to register and generate their encryption key
Upload button is disabled	Select a patient who has a registered key
Only valid PDF files are accepted	Choose a real .pdf file
PDF exceeds the size limit	Use a file smaller than 12 MB

4. Administrator portal

Sign in with the administrator account (demo: admin@securelab.local). The sidebar has three pages: **Overview**, **Staff accounts** and **Security audit**. Administrators manage people and monitor activity; they cannot decrypt patient reports.

4.1 Overview

Four counters (Patients, Laboratory staff, Encrypted reports, Audit events) and a table of the six most recent security events.

4.2 Staff accounts

1. Open **Staff accounts**.
2. Under **New staff member** enter the full name, email address and a temporary password of **12 or more characters**.
3. Choose the **Role**: Laboratory staff or Administrator.
4. Click **Create staff account**. The person appears in the **Staff directory** on the right (use the refresh icon if needed).
5. Give the new person their email and temporary password.

> **Tip:** Patients are not created here. They register themselves from the login screen.

4.3 Security audit

A table of up to 300 recent events. Columns: **Event**, **Actor** (who), **Details**, **IP address** and **Time**.

Event name	Meaning
LOGIN SUCCESS	Someone signed in (look for repeated failures or odd hours)
REPORT ENCRYPTED AND UPLOADED	Lab staff uploaded a report for a patient
ENCRYPTED REPORT ACCESSED	A patient downloaded the encrypted data to open a report
Key registration / MFA events	A patient registered a key or changed two-factor settings

Audit logs are stored in the application database; a production system would send them to a tamper-proof logging service.

5. Full demonstration walk-through

1. **Patient A:** register, generate the key (save the file), optionally enable two-factor, sign out.
2. **Lab:** sign in, upload the sample PDF for Patient A, check **Uploaded reports**, sign out.
3. **Patient A:** sign in, unlock the key file, click **Decrypt & verify report**. The PDF opens.
4. **Admin:** sign in, open **Security audit** and find the login, upload and access events.
5. **Patient B (optional):** register a second patient to show they cannot see Patient A's report (the server returns HTTP 403).

6. Quick troubleshooting

Problem	Fix
Page will not load	Make sure both terminals are running; open http://127.0.0.1:5173
Login fails	Check the email and password; wait a minute if you tried too many times (rate limit)
Signed out after refresh	Normal: sessions are kept in memory only. Sign in again and re-unlock your key
Wrong passphrase / cannot unlock	Re-enter the exact passphrase; make sure you chose the right .securelabkey file
Report fails verification	The data was altered or the key does not match; ask the lab to re-upload

Academic demo - use synthetic data only